

Module 7: Molecular Genetics — Study Questions

1. What are the three components of a nucleotide?
2. Name the four nitrogenous bases in DNA. Which are purines and which are pyrimidines?
3. State the base pairing rules for DNA. How many hydrogen bonds form between A-T? Between G-C?
4. What does it mean that DNA strands are antiparallel?
5. Why must DNA be replicated before a cell divides? What does "semi-conservative" mean?
6. Describe the roles of helicase, DNA polymerase, primase, and ligase in DNA replication.
7. Why can DNA polymerase only build in the $5' \rightarrow 3'$ direction? How does this create the leading and lagging strands?
8. If one strand of DNA reads $5'$ -ATCGGATC- $3'$, write the complementary strand with correct directionality.
9. What is transcription? Where does it occur in eukaryotic cells?
10. Which enzyme performs transcription? What DNA sequences signal where it starts and stops?
11. How does RNA differ from DNA? (List at least three differences.)
12. Given the DNA template strand $3'$ -TACGGCAT- $5'$, write the resulting mRNA sequence.
13. What are the three types of RNA (mRNA, tRNA, rRNA)? Briefly describe the function of each.

14. What post-transcriptional modifications occur in eukaryotes? (5' cap, poly-A tail, intron splicing)
15. What is translation? Describe the roles of the three ribosome sites (A, P, E).
16. What is a codon? What is the start codon? What are the three stop codons?
17. Starting with the DNA template strand 3'-TACAAAGCAATT-5', write the mRNA and determine the amino acid sequence using a codon table.
18. If a single base is inserted into the middle of a gene, what happens to the protein? Why is this called a "frameshift" mutation?
19. A mutation changes one codon from GAA to GAG. Does this change the amino acid? What type of mutation is this?
20. Compare DNA replication, transcription, and translation: location, enzyme, template, product, and direction.